

Target (TGT) Risk MEMO

To: Investment Committee

From: Team Hemisphere

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Subject: Market Risk Audit & Volatility Outlook: TGT

I. Executive Summary

Quantitative modeling of Target Corp (TGT) historical equity returns (2020–2025) indicates an impending, systematic escalation in market risk over the immediate tactical horizon. A structural grid search optimization reveals that Target's mean log returns have decayed into a pure random walk ARIMA(0,0,0), highlighting that directional forecasting yields zero statistical edge. Concurrently, parametric variance modeling using a fat-tailed GARCH (1,1)-Student's t engine detects a strong volatility clustering regime. Forward projections show daily conditional volatility rising from 1.74% to 1.95% over the next 5 trading days, compounding to a 4.15% cumulative variance pool. Risk thresholds are breached; immediate capital preservation measures are required.

II. Historical Market Context

The data asset spanning 2020 through 2025 incorporates two separate macroeconomic regimes: the volatile pandemic supply-chain expansion and the subsequent inflationary consumer compression. Visually and statistically, TGT displays extreme heteroskedasticity (volatility clustering), characterized by massive, isolated variance shocks most notably the historic May 2022 single-day earnings gap contraction of -25%

An Augmented Dickey-Fuller (ADF) test confirmed structural stationarity of the first-differenced log return series ($p < 0.01$), making it eligible for secondary parametric evaluation. However, the sample exhibits a extreme Jarque-Bera statistic fueled by an excess kurtosis of 36.51 and a negative skewness of -2.19. This mathematically proves that standard normal distributions collapse when evaluating Target's tail risks, requiring a heavy-tailed error configuration to avoid systemic under-hedging.

III. Model Forecast Results

A. Mean Return Specification (ARIMA)

An exhaustive grid-search testing 72 permutations of Autoregressive (p), Integrated (d), and Moving Average (q) parameters optimized against the Akaike Information Criterion (AIC) selected a baseline **ARIMA (0,0,0)** (AIC = -5907.32).

- The estimated constant mean drift coefficient ($\mu = 0.0001$, $p = 0.838$) is statistically indistinguishable from zero.
- The **Ljung-Box (Q)** test returns a diagnostic p-value of **0.33**, confirming that all sequential dependencies have been exhausted and residuals represent absolute white noise. Consequently, the 10-day point asset forecast collapses into a flat horizontal mean trajectory.

B. Risk Variance Specification (GARCH)

To accurately track and project variance momentum, a GARCH (1,1) framework was executed using a standardized Student's t-distribution (AIC = 5219.24). The parameter estimation breaks down as follows:

- **Omega ($\omega = 0.7135$, $p = 0.084$):** Baseline invariant long-run variance foundation.
- **Alpha ($\alpha_1 = 0.1248$, $p = 0.102$):** Direct sensitivity to immediate market shocks/news.
- **Beta ($\beta_1 = 0.7196$, $p < 0.001$):** Memory coefficient tracking historical conditional variance.

$$\text{Systemic Volatility Persistence } (\alpha_1 + \beta_1) = 0.8444$$

This high degree of persistence demonstrates that variance shocks possess a half-life of 4.10 days, forcing conditional volatility to mean-revert gradually toward an elevated structural long-run daily baseline of 2.1413%.

5-Day-Ahead Out-of-Sample Volatility Projection

The forward-looking operational path shows an accelerating accumulation of daily risk:

- **Horizon T+1:** Daily Volatility = **1.7399%**(Annualized Vol: 27.6204%)
- **Horizon T+2:** Daily Volatility = **1.8082%** (Annualized Vol: 28.7049%)
- **Horizon T+3:** Daily Volatility = **1.8640%** (Annualized Vol: 29.5897%)
- **Horizon T+4:** Daily Volatility = **1.9098%** (Annualized Vol: 30.3168%)
- **Horizon T+5:** Daily Volatility = **1.9476%** (Annualized Vol: 30.9174%)
- **Total Cumulative 5-Day Exposure Window Volatility: 4.1487%**

IV. Strategic Recommendation

1. **Delta-Neutral Hedging:** Because the mean return is a zero-drift random walk ARIMA (0,0,0) and variance is expanding via GARCH, directional long positions carry negative risk-adjusted utility. Traders should hedge long delta exposure immediately via the options market.
2. **Volatility Positioning (Long Gamma/Vega):** Given the projected expansion of daily volatility toward its 2.14% long-run target, current implied volatility options structures are underpricing tail risk. We recommend executing near-the-money long straddles or strangles to harvest gains from expanding variance.
3. **De-leveraging Tight Stop Boundaries:** The forecast shows annualized asset volatility rising past 30.9%. Quantitative risk managers must widen intraday stop-loss bands to prevent forced liquidation from noise or decrease aggregate position size by at least 25% to preserve capital.